

Vector recovery from saturated frame coefficients

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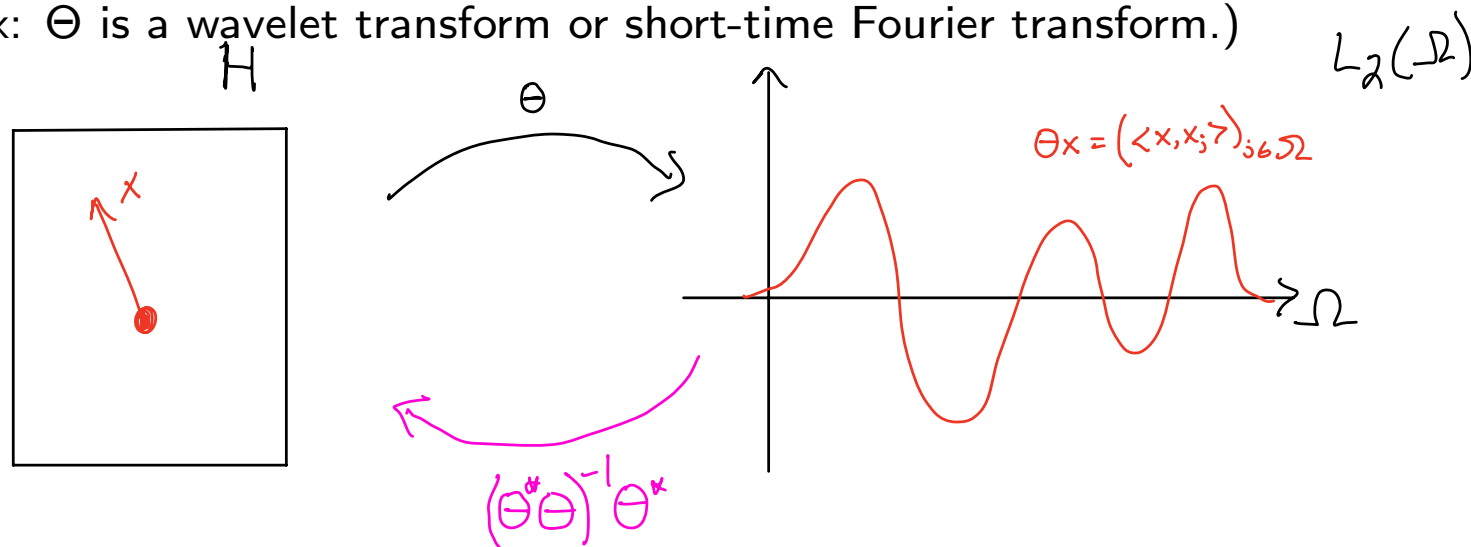
Continuous frames and vector reconstruction

Consider a Hilbert space H and $(x_t)_{t \in \Omega} \subseteq H$ where (Ω, μ) is a measure space.

We want the *analysis operator* $\Theta : H \rightarrow L_2(\Omega)$ to be defined by

$$\Theta(x) = (\langle x, x_t \rangle)_{t \in \Omega} \quad \text{for all } x \in H.$$

(Ex: Θ is a wavelet transform or short-time Fourier transform.)



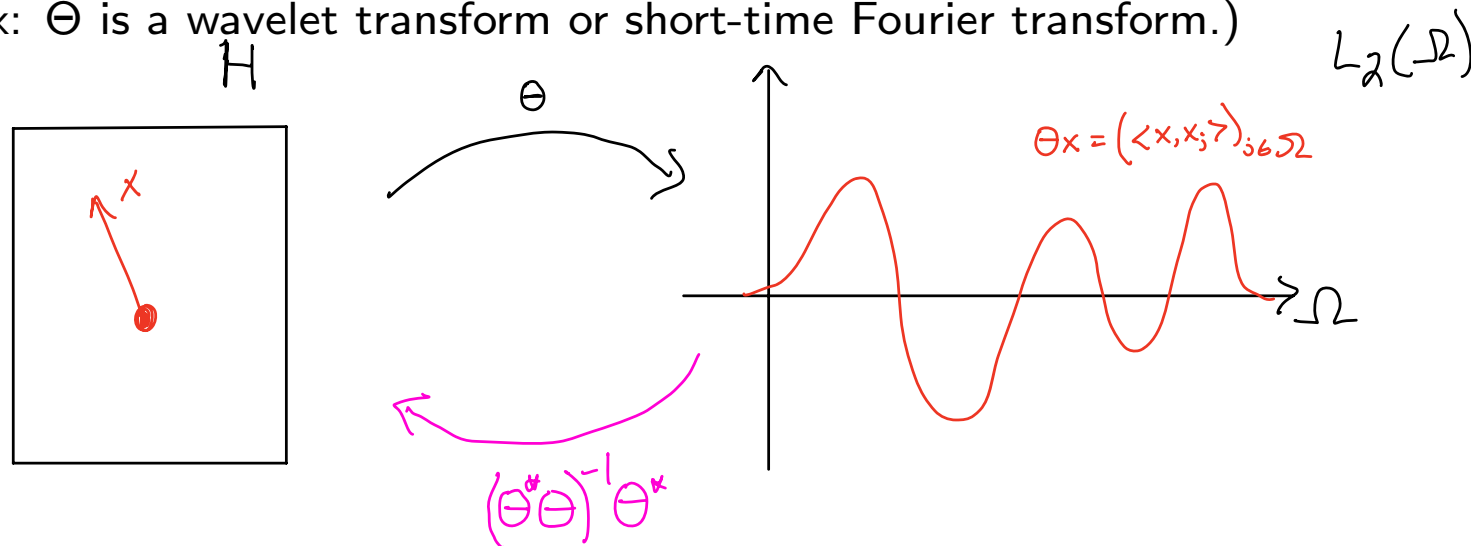
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For both $x \mapsto \Theta x$ and $\Theta x \mapsto x$ to be continuous we need constants $0 < A \leq B$ so that

$$A\|x\|^2 \leq \|\Theta x\|^2 \leq B\|x\|^2 \quad \text{for all } x \in H. \quad (1)$$

That is,

$$A\|x\|^2 \leq \int_{\Omega} |\langle x, x_t \rangle|^2 d\mu(t) \leq B\|x\|^2 \quad \text{for all } x \in H. \quad (2)$$

If (2) holds then we say $(x_t)_{t \in \Omega} \subseteq H$ is a *continuous frame* of H .

If Ω is a discrete set and μ is counting measure, then $(x_t)_{t \in \Omega} \subseteq H$ is a *frame* of H .

Vector reconstruction after non-linear loss of information

Let $(x_t)_{t \in \Omega}$ is a continuous frame of a Hilbert space H .

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Non-linear loss of information:

Suppose $\Phi : L_2(\Omega) \rightarrow L_2(\Omega)$ is both non-linear and not one-to-one.

When and how can we reconstruct x from the measurements $\Phi \Theta x$?

This clearly depends a lot on the non-linear function Φ !

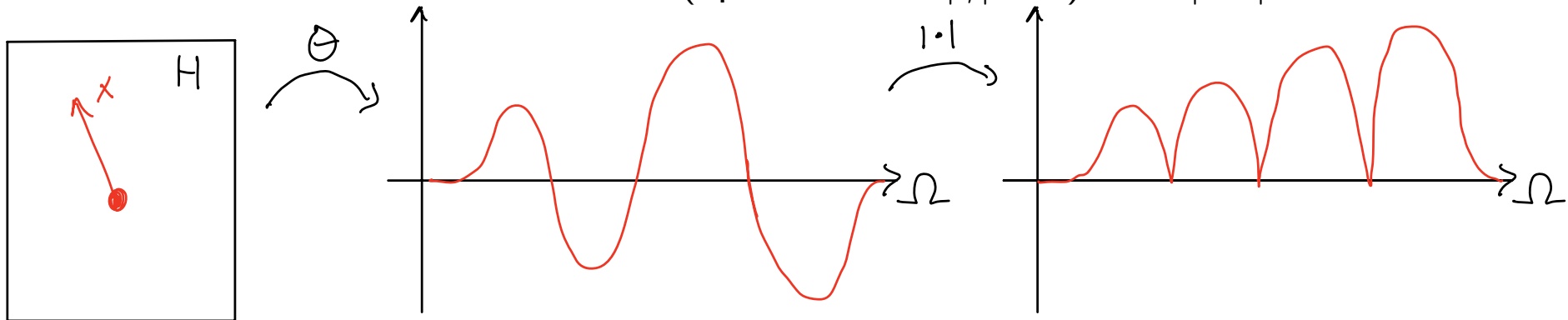
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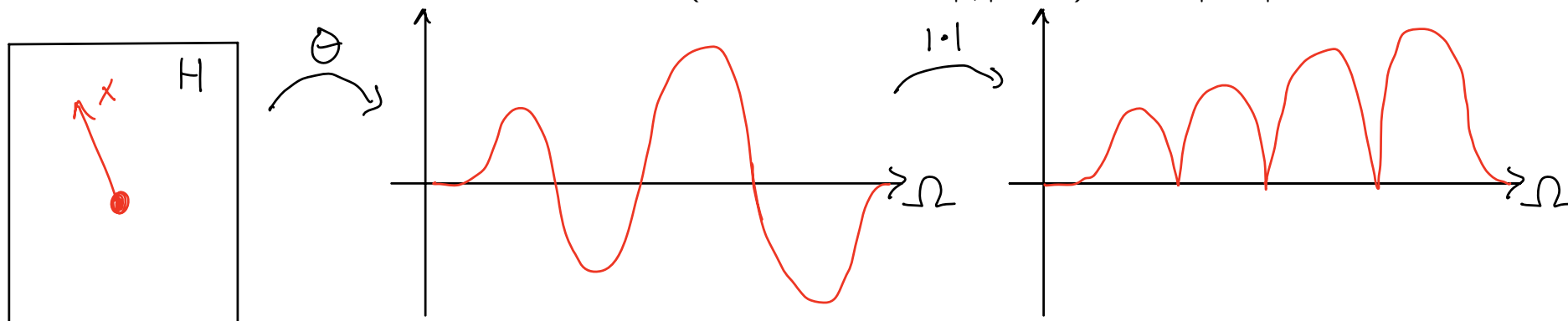
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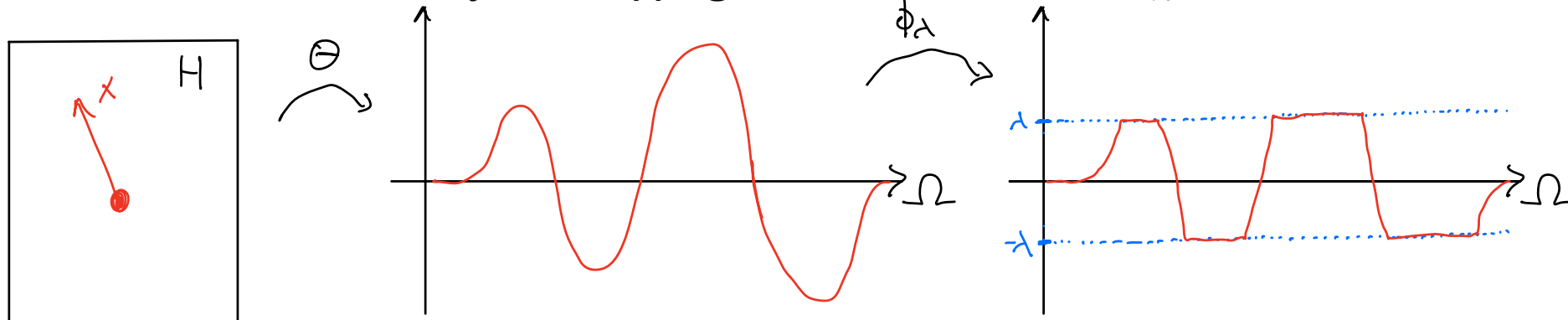
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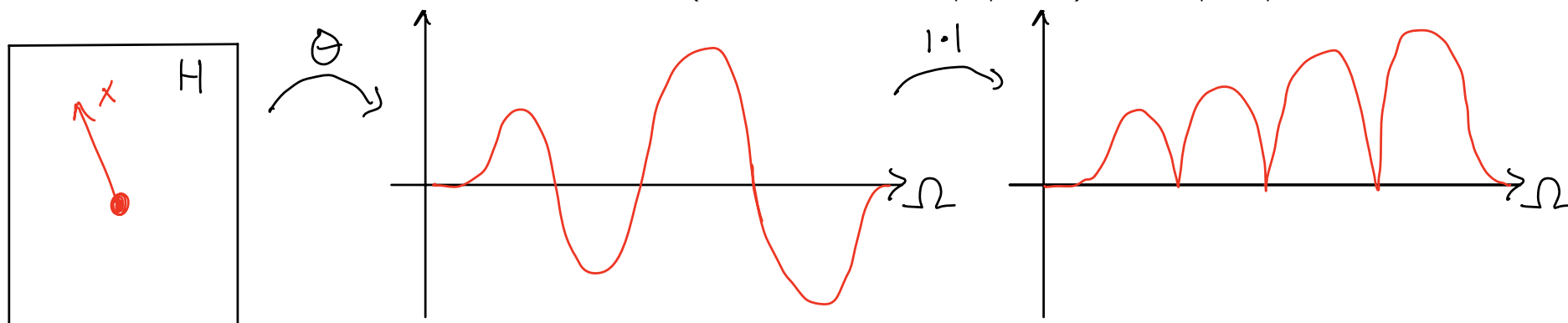
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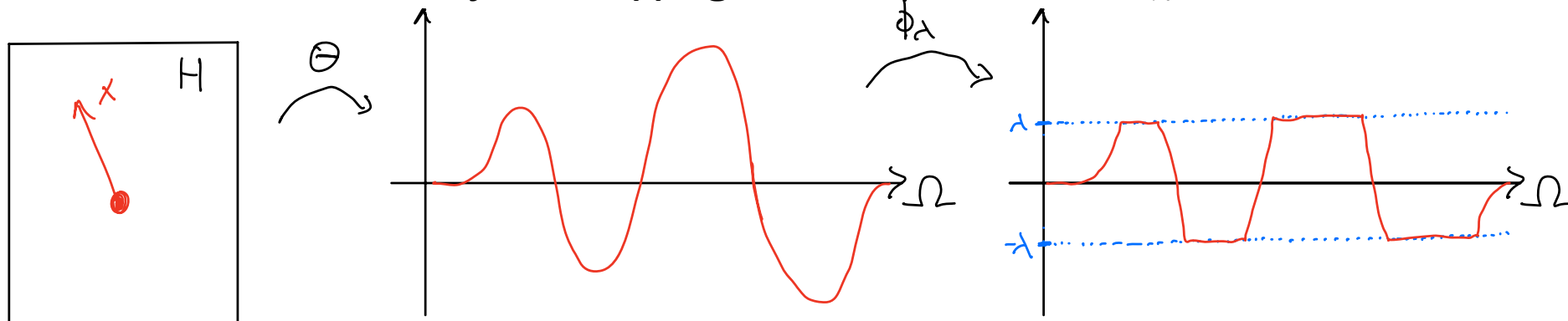
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3. ReLU: (Haider, Ehler, Balazs '23)

4. Mod: (Bhandari, Krahmer, Raskar '17)

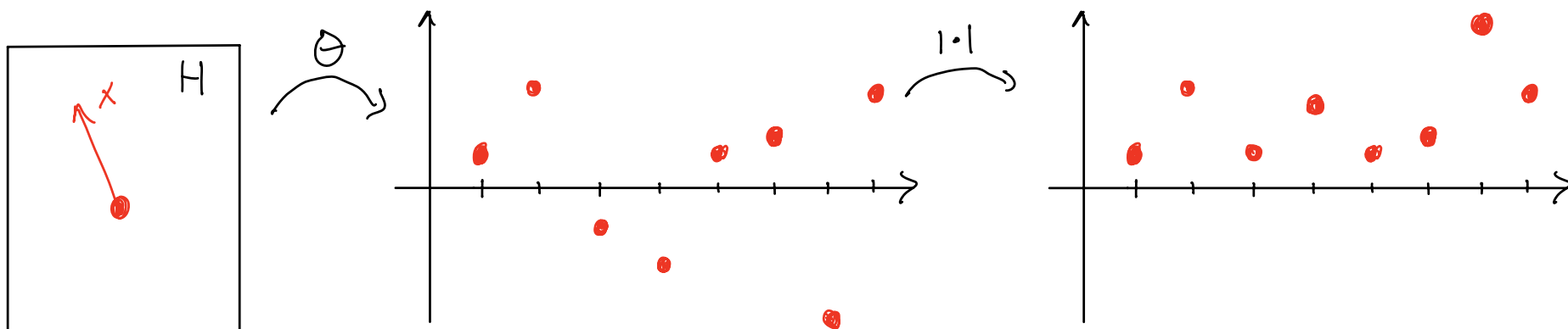
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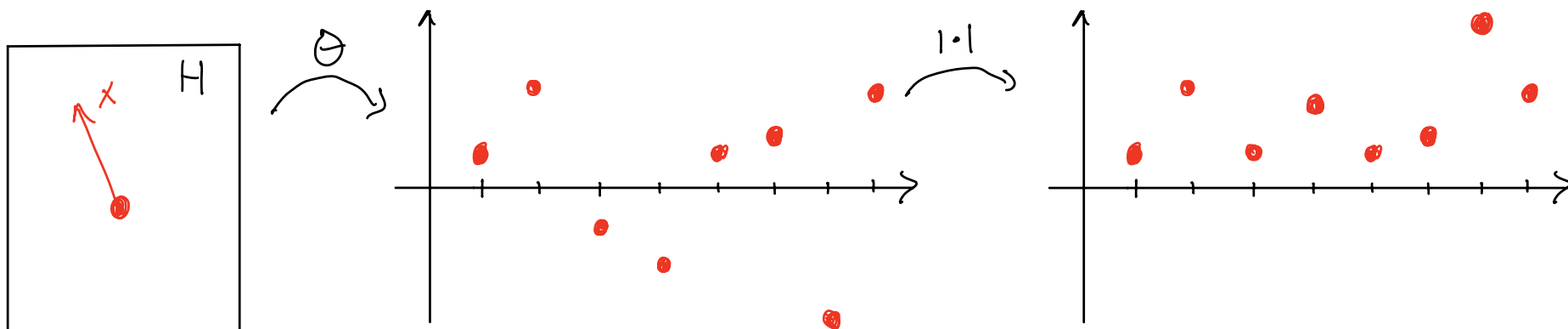
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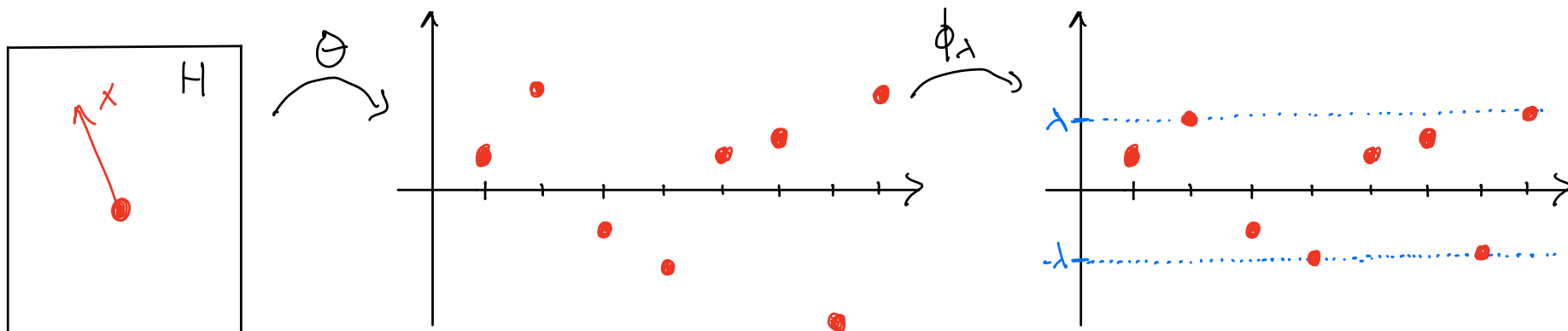
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$(x_j)_{j \in J}$ is a frame of a Hilbert space H with analysis operator $\Theta : H \rightarrow \ell_2(J)$.

Phase retrieval

We define an equivalence relation on H by $x \sim y$ if $x = \gamma y$ for some scalar $|\gamma| = 1$.

We say that $(x_j)_{j \in J}$ does *phase retrieval* on H if $|\Theta| : H/\sim \rightarrow \ell_2(J)$ is one-to-one.

That is, $x = \gamma y$ for some $|\gamma| = 1$ if and only if $|\langle x, x_j \rangle| = |\langle y, x_j \rangle|$ for all $j \in J$.

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Saturation recovery

For $\lambda > 0$, we denote

$$\phi_\lambda(t) = \begin{cases} \lambda & \text{if } t > \lambda, \\ t & \text{if } -\lambda \leq t \leq \lambda \\ -\lambda & \text{if } t < -\lambda \end{cases}$$

We say that $(x_j)_{j \in J}$ does λ -saturation recovery on the unit ball B_H if

$\phi_\lambda \Theta : B_H \rightarrow \ell_2(J)$ is one-to-one.

That is, for all $x, y \in B_H$, $x = y$ if and only if $\phi_\lambda(\langle x, x_j \rangle) = \phi_\lambda(\langle y, x_j \rangle)$ for all $j \in J$.

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Theorem (Balan, Casazza, Edidin '06)

A frame $(x_j)_{j \in J}$ for $\mathbb{R}^n$ does phase retrieval $\Leftrightarrow (x_j)_{j \in J}$ has the complement property.

That is, $(x_j)_{j \in J}$ does phase retrieval if and only if for all $J_0 \subseteq J$ either $(x_j)_{j \in J_0}$ or $(x_j)_{j \in J_0^c}$ is a frame of $\mathbb{R}^n$.

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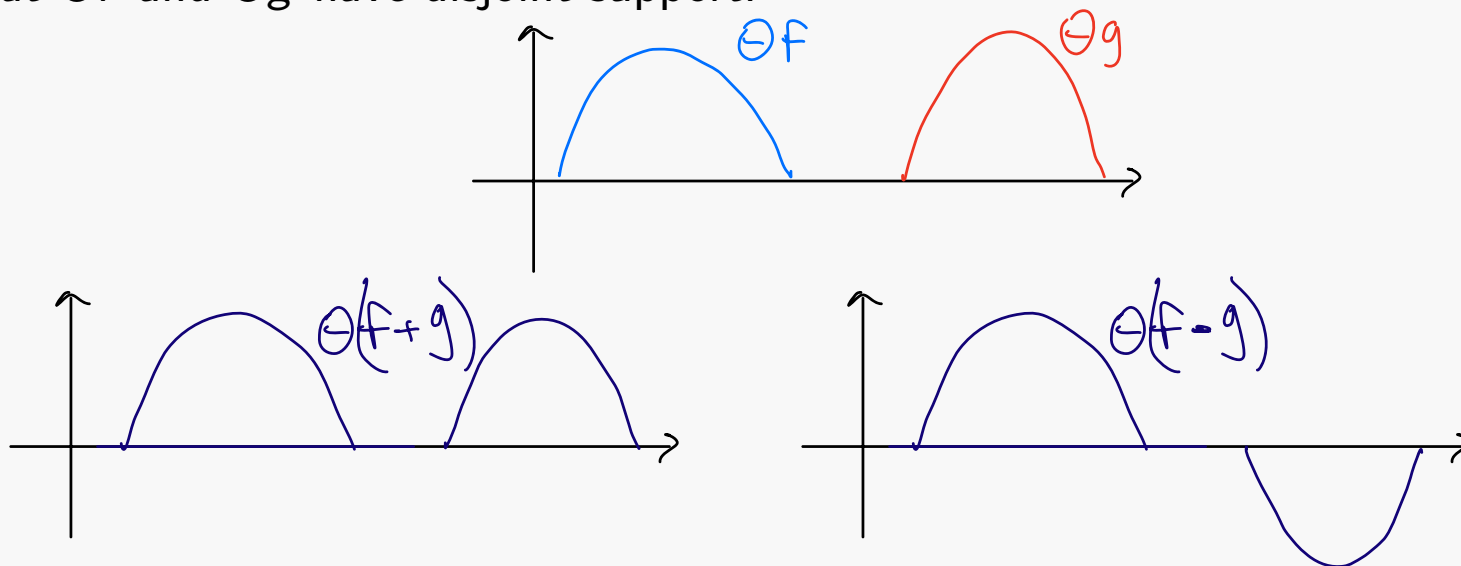
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proof of phase retrieval $\Rightarrow$ complement property.

If $(x_j)_{j \in J}$ fails the complement property then we can find unit vectors $f, g \in \mathbb{R}^n$ such that Θf and Θg have disjoint support.



Thus, $|\Theta(f+g)| = |\Theta(f-g)|$.

However, $f+g \neq f-g$ and $f+g \neq -(f-g)$. So $(x_j)_{j \in J}$ fails phase retrieval. $\square$

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Let $(x_j)_{j \in J}$ be a frame for $\mathbb{R}^n$. For each $x \in B_{\mathbb{R}^n}$, denote

$$J_{\leq \lambda}(x) = \{j \in J : |\langle x, x_j \rangle| \leq \lambda\}.$$

Then $(x_j)_{j \in J}$ does λ -saturation recovery on B_H if and only if for all $x \in B_{\mathbb{R}^n}$, $(x_j)_{j \in J_{\leq \lambda}(x)}$ is a frame of H .

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$(x_j)_{j \in J}$ does phase retrieval for $\mathbb{R}^n \iff$

Whenever $(x_j)_{j \in J}$ is partitioned into two sets, at least one of the sets is a frame.

$(x_j)_{j \in J}$ does λ -saturation recovery for $B_{\mathbb{R}^n} \iff$

Whenever $(x_j)_{j \in J}$ is partitioned into the saturated coordinates and the unsaturated coordinates, the unsaturated coordinates form a frame.

In [Haider, Ehler, Balazs '23], the analogous theorem is proven for using the ReLU function instead of the saturation function ϕ_λ .

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Corollary (Balan, Casazza, Edidin '06)

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Cases:

For $m = n$: Every basis of unit vectors for $\mathbb{R}^n$ does 1-saturation recovery on $B_{\mathbb{R}^n}$, and if $1 > \lambda > 0$ then no basis of unit vectors for $\mathbb{R}^n$ does λ -saturation recovery on $B_{\mathbb{R}^n}$.

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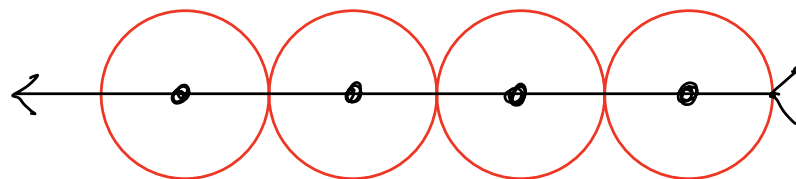
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For $m = n + 1$: An equi-angular frame of $n + 1$ unit vectors for $\mathbb{R}^n$ does $2^{-1/2}(1 + 1/n)^{1/2}$ -saturation recovery on $B_{\mathbb{R}^n}$, and if $2^{-1/2}(1 + 1/n)^{1/2} > \lambda > 0$ then no frame of $n + 1$ unit vectors for $\mathbb{R}^n$ does λ -saturation recovery on $B_{\mathbb{R}^n}$.

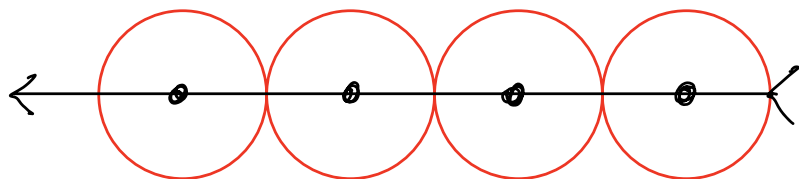
Brief excursion into sphere packing

Let (X, d) be a metric space. For $x_0 \in X$ and $\varepsilon > 0$ we denote $B_\varepsilon^o(x_0) = \{x \in X : d(x, x_0) < \varepsilon\}$ to be the open ball of radius ε centered at x_0 . A *packing* in X of balls of radius ε is a collection of disjoint balls $(B_\varepsilon^o(x_j))_{j=1}^m$.

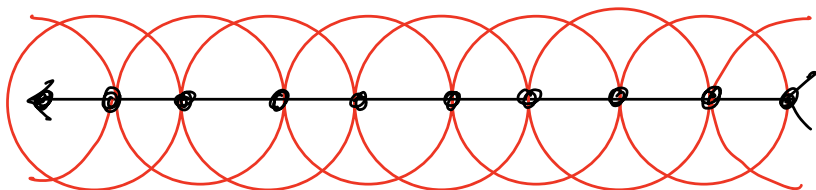


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Given $k \in \mathbb{N}$, we call $(B_\varepsilon^o(x_j))_{j=1}^m$ a *k-fold multi packing* or *k-fold multiple packing* if for all $x \in X$ we have that $d(x, x_j) < \varepsilon$ for at most k values of $j \in 1, \dots, m$.



Note that $(B_\varepsilon^o(x_j))_{j=1}^m$ is a packing in X if and only if it is a 1-fold multi packing.

Packings in Projective space

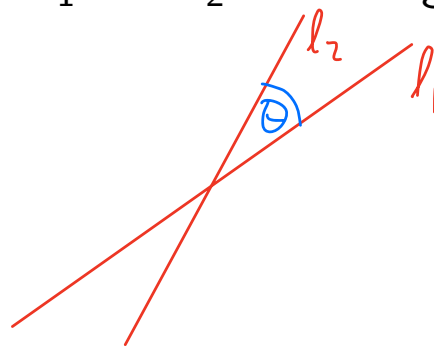
P^{n-1} is the set of lines through the origin in $\mathbb{R}^n$.

The distance between lines ℓ_1 and ℓ_2 is the angle $\theta \in [0, \pi/2)$ between them.

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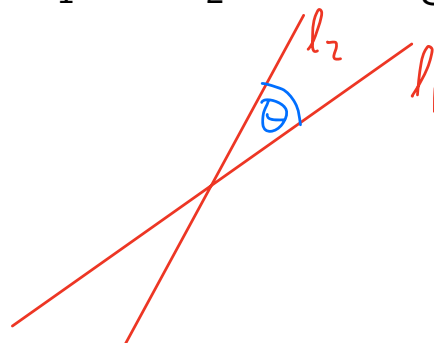
For $x \in \mathbb{R}^n$ with $x \neq 0$, we denote $[x] = \text{span}(x)$ to be the line containing x .

Then for $x, y \in \mathbb{R}^n$ with $\|x\| = \|y\| = 1$ we have that $d([x], [y]) = \cos^{-1}(|\langle x, y \rangle|)$

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Huge problems: Given $m \geq n$, what is the least $\varepsilon > 0$ so that there exists unit vectors

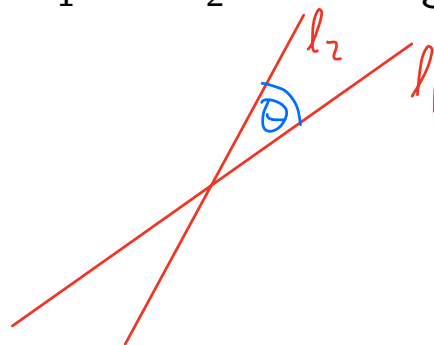
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How can we construct an optimal choice for $(x_j)_{j=1}^m \subseteq \mathbb{R}^n$?

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Welch bound: If $(x_j)_{j=1}^m \subseteq S_{\mathbb{R}^n}$ are unit vectors and $m > n$ then

$$\max_{k \neq j} |\langle x_k, x_j \rangle| \geq \left(\frac{m-n}{n(m-1)} \right)^{1/2}$$

Theorem (Strohmer, Heath '03)

Let $(x_j)_{j=1}^m \subseteq S_{\mathbb{R}^n}$ be unit vectors and $m > n$. Then equality is achieved in the Welch bound if and only if $(x_j)_{j=1}^m$ is an equiangular tight frame.

Multi-Packings in Projective space

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A frame $(x_j)_{j=1}^m$ for $\mathbb{R}^n$ is called *full spark* if every subset of n vectors is a basis of $\mathbb{R}^n$.

Corollary (Alharbi, F, Ghoreishi, Johnson, Randrianarivony '23)

Let $(x_j)_{j=1}^m$ be a full spark frame of unit vectors for $\mathbb{R}^n$ and let $0 < \lambda < 1$. Then $(x_j)_{j=1}^m$ does λ -saturation recovery on $B_{\mathbb{R}^n}$ if and only if $(B_\varepsilon([x_j]))_{j=1}^m$ is an $(m - n)$ -fold multi packing in P^{n-1} for $\varepsilon = \cos^{-1}(\lambda)$.

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Proof.

$$\begin{aligned} (x_j)_{j=1}^m \text{ does } \lambda\text{-sat rec. on } B_{\mathbb{R}^n} &\Leftrightarrow (x_j)_{j \in J_{\leq \lambda}(x)} \text{ is a frame for all } x \in S_{\mathbb{R}^n}, \\ &\Leftrightarrow |\{j : |\langle x, x_j \rangle| \leq \lambda\}| \geq n \text{ for all } x \in S_{\mathbb{R}^n}, \\ &\Leftrightarrow |\{j : |\langle x, x_j \rangle| > \lambda\}| \leq m - n \text{ for all } x \in S_{\mathbb{R}^n}, \\ &\Leftrightarrow |\{j : d([x], [x_j]) < \cos^{-1}(\lambda)\}| \leq m - n \text{ for all } [x] \in P^{n-1}, \\ &\Leftrightarrow (B_{\cos^{-1}(\lambda)}([x_j]))_{j=1}^m \text{ is an } (m - n)\text{-fold multi packing in } P^{n-1}. \end{aligned}$$

□

Packing problems in projective space

Hard problems that lots of people have worked on:

Given $m > n$, what is the least $\varepsilon > 0$ so that there exists unit vectors $(x_j)_{j=1}^m \subseteq S_{\mathbb{R}^n}$ such that $(B_\varepsilon([x_j]))_{j=1}^m$ is a packing in P^{n-1} ?

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Hard problems that much less people have worked on:

Given $m > n$ and $k \geq 1$, what is the least $\varepsilon > 0$ so that there exists unit vectors $(x_j)_{j=1}^m \subseteq S_{\mathbb{R}^n}$ such that $(B_\varepsilon([x_j]))_{j=1}^m$ is a k -fold multiple packing in P^{n-1} ?

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Phase retrieval

Suppose that $(x_j)_{j \in J}$ does phase retrieval for H .

This gives a well defined recovery map $|\Theta x| \mapsto x$ from the range of $|\Theta|$ to $H/\sim$.

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We say that $(x_j)_{j \in J}$ does **C-stable phase retrieval** if $|\Theta x| \mapsto x$ is C-Lipschitz.

That is,

$$\min_{|\lambda|=1} \|x - \lambda y\| \leq C \| |\Theta x| - |\Theta y| \| \quad \text{for all } x, y \in H.$$

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Let $(x_j)_{j=1}^m \subseteq \mathbb{R}^n$ be a finite frame and let

$\lambda_c = \inf\{\lambda : (x_j)_{j=1}^m \text{ does } \lambda\text{-saturation recovery on } B_{\mathbb{R}^n}\}.$

Then $(x_j)_{j=1}^m$ does λ_c -saturation recovery on $B_{\mathbb{R}^n}$. Furthermore, for all $\lambda > \lambda_c$ we have that $(x_j)_{j=1}^m$ does stable λ -saturation recovery on $B_{\mathbb{R}^n}$.

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Question: Does $(x_j)_{j=1}^m$ do stable λ_c -saturation recovery on $B_{\mathbb{R}^n}$?

For each $\lambda > 0$, it is possible to construct a frame $(x_j)_{j=1}^\infty$ of unit vectors for ℓ_2 which does stable λ -saturation recovery on B_{ℓ_2} .

Question: Does there exist a frame $(x_j)_{j=1}^\infty$ of unit vectors for ℓ_2 which does stable λ -saturation recovery on B_{ℓ_2} for all $\lambda > 0$?

Constructing frames which do C -stable phase retrieval

If $m \geq 2n - 1$ then almost every frame $(x_j)_{j=1}^m$ of $\mathbb{R}^n$ does phase retrieval. However, the stability constant for phase retrieval could be very bad.

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How can we construct frames which do C -stable phase retrieval for $\mathbb{R}^n$ where C is independent of the dimension n ?

Theorem (E. J. Candès, T. Strohmer, and V. Voroninski (2013))

There exists a constant $C > 0$ such that if $(x_j)_{j=1}^m$ are chosen randomly and independently with uniform distribution on the unit sphere in $\mathbb{R}^n$ and m is chosen on the order of $n \log(n)$ then with high probability $(x_j)_{j=1}^m$ does C -stable phase retrieval.

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There exists a constant $C > 0$ such that if $(x_j)_{j=1}^m$ are chosen randomly and independently with uniform distribution on the unit sphere or Gaussian distribution in $\mathbb{R}^n$ and m is chosen on the order of n then with high probability $(x_j)_{j=1}^m$ does C -stable phase retrieval.

[F. Krahmer and Y. Liu (2018), F. Krahmer and D. Stöger (2020)] extend the above result to a large number of different sub-Gaussian distributions.

Discretizing phase retrieval

Can continuous frames which do stable phase retrieval for $\mathbb{R}^n$ be discretized using m on the order of n sampling points to obtain a frame which does stable phase retrieval?

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Theorem (F, D. Ghoreishi '22)

Let $(x_t)_{t \in \Omega}$ be a continuous Parseval frame for $\mathbb{R}^n$ over a probability space Ω which does κ -stable phase retrieval and $\|x_t\| \leq \beta\sqrt{n}$ for all $t \in \Omega$. Let $\Theta : \ell_2^n \rightarrow L_2(\Omega)$ be the analysis operator. Suppose that $(\frac{1}{\sqrt{m}}x_{t_j})_{j=1}^m$ is a frame of $\mathbb{R}^n$ with upper frame bound B and lower frame bound A which does C -stable phase retrieval. Then both the L_2 norm and the L_1 norm on the range of the analysis operator are discretized in the following way for all $x \in \mathbb{R}^n$,

1

$$A\|\Theta x\|_{L_2(\Omega)}^2 \leq \frac{1}{m} \sum_{j=1}^m |\langle x, x_{t_j} \rangle|^2 \leq B\|\Theta x\|_{L_2(\Omega)}^2$$

2

$$\frac{A^{1/2}}{B^{3/2}C^3(1+A^{-1}\beta^2)^{3/2}}\|\Theta x\|_{L_1(\Omega)} \leq \frac{1}{m} \sum_{j=1}^m |\langle x, x_{t_j} \rangle| \leq B^{1/2}\kappa^3(1+\beta^2)^{3/2}\|\Theta x\|_{L_1(\Omega)}$$

Constructing frames which do C -stable λ -saturation recovery

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Let v be a random vector which is uniformly distributed in $S_{\mathbb{R}^n}$. Then,

$$\mathbb{E}|\langle x, v \rangle|^2 = \frac{1}{n} \quad \text{for all } x \in S_{\mathbb{R}^n}.$$

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Theorem (J. Laska, P. Boufounos, M. Davenport, R. Baraniuk '13
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There exists $1 > \alpha > 0$ and a constant $C_\alpha > 0$ such that for all $n \in \mathbb{N}$ if $\lambda \geq \alpha/\sqrt{n}$ and $(x_j)_{j=1}^m$ are chosen randomly and independently with uniform distribution in $S_{\mathbb{R}^n}$ and m is chosen on the order of n then with high probability

$$\|x - y\|^2 \leq C_\alpha^2 \frac{n}{m} \sum_{j=1}^m |\phi_\lambda(\langle x, x_j \rangle) - \phi_\lambda(\langle y, x_j \rangle)|^2 \quad \text{for all } x, y \in S_{\mathbb{R}^n}.$$

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For all $\alpha > 0$ there exists $C_\alpha > 0$ such that for all $n \in \mathbb{N}$ if $\lambda \geq \alpha/\sqrt{n}$ and $(x_j)_{j=1}^m$ are chosen randomly and independently with uniform distribution in $S_{\mathbb{R}^n}$ and m is chosen on the order of $n \log(n)$ then with high probability

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1. Show that there exists $A_1 > 0$ so that

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2. Use concentration inequalities to prove that for each $x, y \in S_{\mathbb{R}^n}$ if $(v_j)_{j=1}^m$ are iid copies of v and m is on the order of n then with high probability

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5. If α is sufficiently close to 1 then ε can be chosen to be a constant, and m can be chosen on the order of n .

Otherwise, ε can be chosen on the order $n^{-1/2}$, and m on the order of $n \log(n)$.

Implementing λ -saturation recovery

$(x_j)_{j \in J}$ is a frame of H with analysis operator $\Theta : H \rightarrow \ell_2(J)$.

General problem: We obtain measurements $(b_j)_{j \in J} = \phi_\lambda \Theta x = (\phi_\lambda(\langle x, x_j \rangle))_{j \in J}$.

Our goal is to reconstruct the vector $x \in H$.

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That is, we need to solve the following for $y \in H$:

$$\begin{aligned} \langle y, x_j \rangle &\geq \lambda && \text{if } b_j = \lambda, \\ \langle y, x_j \rangle &= b_j && \text{if } -\lambda < b_j < \lambda \\ \langle y, x_j \rangle &\leq -\lambda && \text{if } b_j = -\lambda \end{aligned}$$

Note: For the case of compressed sensing, we know that $x \in H$ is some k -sparse vector. In [Foucart, Needham '17][Foucart, Li '18], the above conditions are expressed as a linear programming problem and it is shown that when $|J| \geq Ck \ln(en/k)$ and $(x_j)_{j \in J}$ are iid Gaussian random vectors then with high probability we have that ℓ_1 -minimization can be used to recover x .

The frame algorithm (with no saturation)

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Define a sequence $(y_k)_{k=0}^{\infty}$ in H by $y_0 = 0$ and

$$y_{k+1} := y_k + \alpha \Theta^*(\Theta x - \Theta y_k) = y_k + \alpha \sum_{j \in J} \langle x - y_k, x_j \rangle x_j \quad \text{for all } k \geq 0. \quad (4)$$

Then $\|x - y_{k+1}\| \leq C_\alpha \|x - y_k\|$ for all $k \geq 0$, where $C_\alpha := \max\{|1 - \alpha A|, |1 - \alpha B|\}$.

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Then $\|x - y_{k+1}\| \leq C_\alpha \|x - y_k\|$ for all $k \geq 0$, where $C_\alpha := \max\{|1 - \alpha A|, |1 - \alpha B|\}$.

In particular, $(y_k)_{k=0}^\infty$ converges to x and satisfies

$$\|x - y_k\| \leq C_\alpha^k \|x\| \quad \text{for all } k \geq 0.$$

Note that $\frac{B-A}{B+A}$ is the optimal value for C_α and it occurs when $\alpha = \frac{2}{A+B}$.

λ -saturated frame algorithm

$(x_j)_{j \in J}$ is a frame of H with analysis operator $\Theta : H \rightarrow \ell_2(J)$.

General problem: We obtain measurements $(b_j)_{j \in J} = \phi_\lambda \Theta x = (\phi_\lambda(\langle x, x_j \rangle))_{j \in J}$.

The λ -saturated frame algorithm gives a method for approximating the vector $x \in H$.

We denote the following sets,

Unsaturated coordinates: $J_{\leq \lambda}(x) = \{j \in J : |\langle x, x_j \rangle| \leq \lambda\},$

Positively saturated coordinates: $J_{S^+}(x) = \{j \in J : \langle x, x_j \rangle > \lambda\},$

Negatively saturated coordinates: $J_{S^-}(x) = \{j \in J : \langle x, x_j \rangle < -\lambda\}.$

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The λ -saturated frame algorithm gives a method for approximating the vector $x \in H$.

We denote the following sets,

Unsaturated coordinates: $J_{\leq \lambda}(x) = \{j \in J : |\langle x, x_j \rangle| \leq \lambda\},$

Positively saturated coordinates: $J_{S+}(x) = \{j \in J : \langle x, x_j \rangle > \lambda\},$

Negatively saturated coordinates: $J_{S-}(x) = \{j \in J : \langle x, x_j \rangle < -\lambda\}.$

Let $(\alpha_k)_{k=0}^\infty, (\beta_k)_{k=0}^\infty \subseteq [0, \infty)$ be two sequences of scalars.

The sequence $(y_k)_{k=0}^\infty \subseteq H$ is constructed by $y_0 = 0$ and for $k \in \mathbb{N}_0$ we let

$$\begin{aligned} y_{k+1} = y_k + \alpha_k \sum_{j \in J_{\leq \lambda}(x)} \left(\langle x, x_j \rangle - \langle y, x_j \rangle \right) x_j + \beta_k \sum_{j \in J_{S+}(x) \cap J_{S+}^c(y)} \left(\lambda - \langle y_k, x_j \rangle \right) x_j \\ + \beta_k \sum_{j \in J_{S-}(x) \cap J_{S-}^c(y)} \left(-\lambda - \langle y_k, x_j \rangle \right) x_j. \end{aligned}$$

λ -saturated frame algorithm

Let $(x_j)_{j \in J}$ be a frame for a Hilbert space H with frame bounds $A \leq B$. Suppose $(y_k)_{k=0}^\infty \subseteq H$ is constructed by $y_0 = 0$ and for $k \in \mathbb{N}_0$ we let

$$\begin{aligned} y_{k+1} = y_k &+ \frac{2}{A+B} \sum_{j \in J_{\leq \lambda}(x)} \left(\langle x, x_j \rangle - \langle y, x_j \rangle \right) x_j + \frac{2}{A+B} \sum_{j \in J_{S^+}(x) \cap J_{S^+}^c(y)} \left(\lambda - \langle y_k, x_j \rangle \right) x_j \\ &+ \frac{2}{A+B} \sum_{j \in J_{S^-}(x) \cap J_{S^-}^c(y)} \left(-\lambda - \langle y_k, x_j \rangle \right) x_j. \end{aligned}$$

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Suppose that $(x_j)_{j \in J_{\leq \lambda}(x)}$ has optimal lower frame bound $A' > 0$ and that $(x_j)_{j \in J_{\leq \lambda}(x) \cup (J_{S^+}(x) \cap J_{S^+}^c(y)) \cup (J_{S^-}(x) \cap J_{S^-}^c(y))}$ has a lower frame bound A'' with $A'' > A'$.

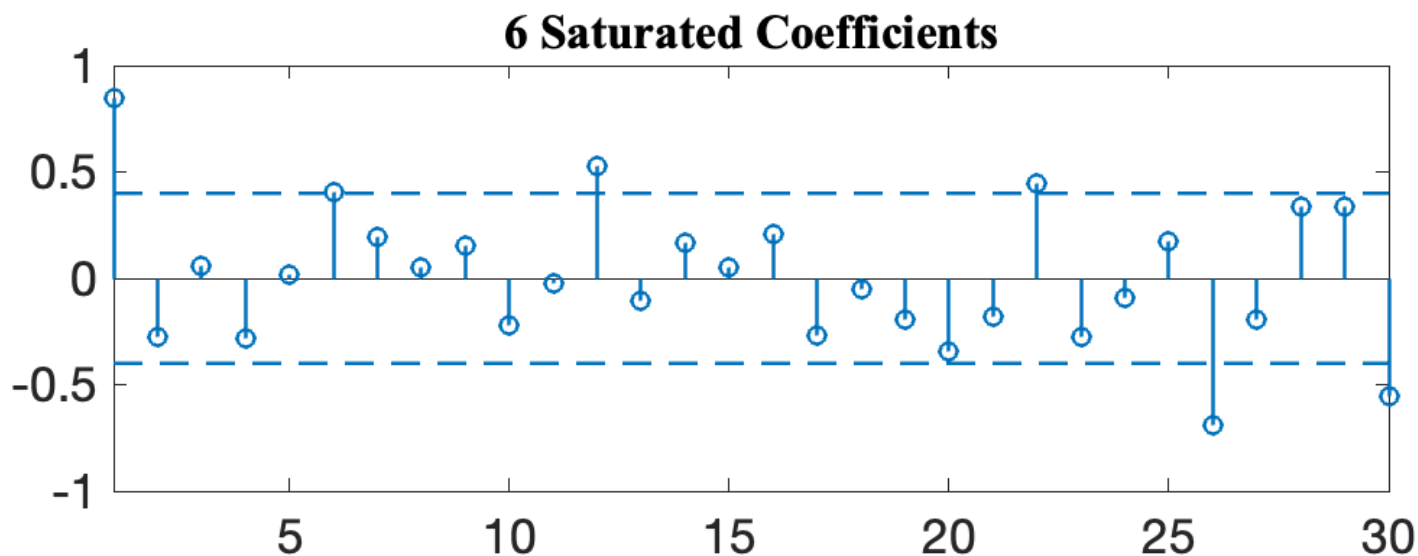
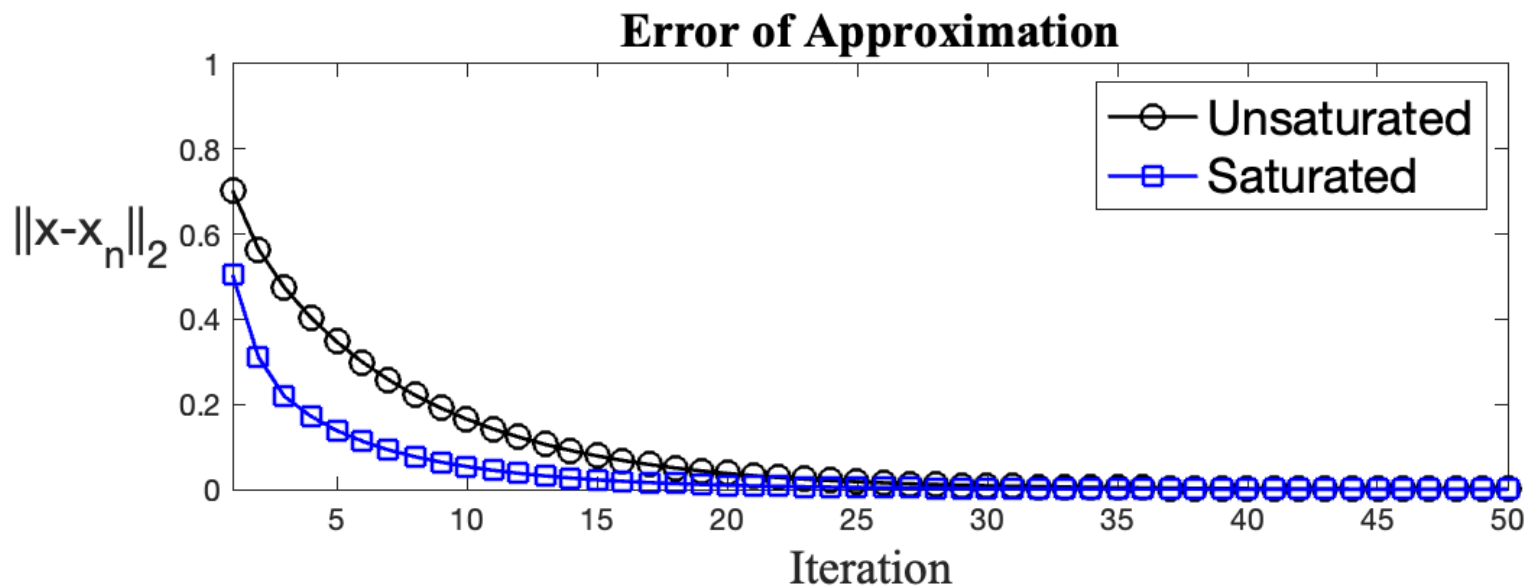
Let $C_{2/(A+B)}$ be the constant for applying the frame algorithm to $(x_j)_{j \in J_{\leq \lambda}(x)}$ using coefficient $\alpha = 2/(A+B)$.

Then there exists $\varepsilon_k > 0$ so that

$$\|x - y_{k+1}\| \leq (1 - \varepsilon_k) C_{2/(A+B)} \|x - y_k\|.$$

Numerical Example: $\lambda = 0.4$ (Low Saturation)

The frame $(x_j)_{j=1}^{30} \subseteq \mathcal{S}_{\mathbb{R}^{10}}$ is chosen randomly and independently with uniform distribution in $\mathcal{S}_{\mathbb{R}^{10}}$.



Numerical Example: $\lambda = 0.25$ (High Saturation)

The frame $(x_j)_{j=1}^{30} \subseteq S_{\mathbb{R}^{10}}$ is chosen randomly and independently with uniform distribution in $S_{\mathbb{R}^{10}}$.

